FitLife Billboard Case

Chronological session log | Verbatim user prompts and summaries of outputs

Transcription note: user prompts are preserved as written in the conversation, including spelling and code. The plot supplied as an image is identified by a descriptive placeholder. Assistant outputs are summarized rather than reproduced verbatim.

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01. User prompt (verbatim)

```
Again, I have a webinar for the coding. I will be using to validate understanding, work
through concepts and errors
```

Summary of output / discussion: Agreed to use the webinar as the coding guide, with assistance focused on conceptual validation, output interpretation, and debugging rather than proactively completing the assignment.

02. User prompt (verbatim)

```
here is the webinar code. I need to start at panel data, yes?
```

Summary of output / discussion: Identified Panel Data at approximately line 209. Earlier ShopHub content is review; the grocery-store simulation illustrates fixed effects. Billboard Case begins around line 436.

03. User prompt (verbatim)

```
help me break out what sections of code belong to what question
```

Summary of output / discussion: Mapped Q1 to pooled regression example (305-313); Q2 to covariate regression and Billboard instructions (461-464); Q3 to 1WFE demonstration (316-370); Q4 to 2WFE (401-410); Q5 to final case prompts (469-475).

04. User prompt (verbatim)

```
what library's do i need
```

Summary of output / discussion: Recommended library(tidyverse) for data manipulation and plots and library(fixest) for feols fixed-effects estimators; lm is included in base R.

05. User prompt (verbatim)

```
okay, remind me which sections go to each question
```

Summary of output / discussion: Repeated the webinar-to-assignment crosswalk and advised beginning with Billboard-specific data setup around lines 440-455.

06. User prompt (verbatim)

```
do I need to prep doing 213-284
```

Summary of output / discussion: Clarified lines 213-284 are conceptual explanation and simulated grocery-store practice rather than required preprocessing of the Billboard data.

07. User prompt (verbatim)

```
# R code and code annotations go within the chunks
lm(downloads ~ tau, data = ds)
```

```

ATE is 3.185 for tau.

Coefficients:
(Intercept)          tau
      6.586        3.185

this answers the middle structure for q1. I have looked through the webinar and my
previous two cases and I am unsure how to code the plot or the confidence. can you help me
 with the code structure?
```

Summary of output / discussion: Distinguished the regression coefficient on tau (3.185) from the true ATT, which requires averaging tau for treated observations in the post period. Supplied tidyverse group_by(date, treated)/summarise and ggplot line structure; used lm(downloads ~ treated) and confint for pooled estimate.

---

08. User prompt (verbatim)

```
lm(downloads ~ treated, data = ds) Why not this?
```

Summary of output / discussion: Confirmed this is precisely the Q1 pooled OLS specification given by the assignment. The treatment coefficient is the pooled treated-control difference, not the known true ATT.

---

09. User prompt (verbatim)

```
so for the plot?
```

Summary of output / discussion: Provided daily_avg <- ds %>% group_by(date, treated) %>% summarise(avg_downloads = mean(downloads, na.rm=TRUE), .groups="drop") and ggplot(... aes(date, avg_downloads, color=factor(treated))) + geom_line() + labs(...). Discussed reading pre/post gaps.

---

10. User prompt (verbatim)

```
Call:
lm(formula = downloads ~ treated, data = ds)

Coefficients:
(Intercept)      treated
      6.444        2.182

               2.5 %   97.5 %
(Intercept) 6.408146 6.480203
treated     2.096623 2.268154

ATE is 2.182. Confidence Interval is [2.09, 2.27]
```

Summary of output / discussion: Validated pooled estimate 2.182, 95% CI [2.097, 2.268] and intercept 6.444. Noted causal interpretation is limited by preexisting differences and the pooled interval uses conventional lm uncertainty.

---

11. User prompt (verbatim)

```
[User shared plot: Daily Average Downloads: Treatment vs. Control; treated group already
higher before May 15 and rises after May 15.]
```

Summary of output / discussion: Graph review: treated cities were roughly 8.0-8.3 pre-campaign, controls about 6.2-6.5; treated series rises to around 9 after campaign while control stays broadly stable. The large pretreatment baseline gap is evidence against attributing the pooled difference entirely to billboards.

---

12. User prompt (verbatim)

```
overestimating
```

Summary of output / discussion: Confirmed the plot suggests pooled OLS overestimates the incremental advertising effect due to the pretreatment gap; encouraged verifying against tau.

---

13. User prompt (verbatim)

```
att is 3.185?
lm(formula = downloads ~ tau, data = ds)

Coefficients:
(Intercept)          tau
      6.586        3.185
```

Summary of output / discussion: Explained again that lm(downloads ~ tau) does not equal the true ATT. Directed calculation: ds %>% filter(treated == 1, post == 1) %>% summarise(ATT = mean(tau, na.rm=TRUE)).

---

14. User prompt (verbatim)

```
> ds %>%
+   filter(treated == 1, post == 1) %>%
+   summarise(ATT = mean(tau, na.rm = TRUE))
# A tibble: 1 x 1
    ATT
  <dbl>
1  0.77

att 0.77?
```

Summary of output / discussion: Confirmed true ATT = 0.770 from tau for treated post observations, compared with pooled 2.182 and the absolute error of 1.412.

---

15. User prompt (verbatim)

```
So, the True ATT is 0.77 additional daily downloads. Pooled ATE is 2.182 with a [2.097,
2.268] 95% Confidence Interval. So the pooled model overestimates the treatment effect by
a little under 2 additional downloads per day
```

Summary of output / discussion: Validated interpretation, refined absolute overestimation to 2.182 - 0.770 = 1.412 downloads per city per day.

---

16. User prompt (verbatim)

```
The new Pooled ATE is 1.37 with a Confidence Interval of [2.09, 2.27]. The new pooled ATE
is closer to the True ATT (0.6 vs orginal 1.412), but the confidence interval remained the
 same. Adding additional covariates helps reduce bias by introducing factors to better
define the groups of people to analyze behaviors for.
```

Summary of output / discussion: Identified stale CI from Q1; clarified Q2 adjusts city-level observed confounders, not groups of people. Q2 coefficient near 1.372; error ~0.602.

---

17. User prompt (verbatim)

```
Call:
lm(formula = downloads ~ treated + city_size + income_level +
    climate + gym_density, data = ds)

Coefficients:
```

```
       (Intercept)             treated     city_sizemedium      city_sizesmall
            7.7726              1.3718             -0.5893             -1.0082
   income_levellow  income_levelmiddle         climatemild         climatewarm
           -0.6228             -0.2844             -0.2431             -0.1177
    gym_densitylow   gym_densitymedium
           -0.4942             -0.2565

                2.5 %   97.5 %
(Intercept) 6.408146 6.480203
treated     2.096623 2.268154
```

Summary of output / discussion: Diagnosed confint() applied to Q1 model rather than the new Q2 model; instructed saving covariate-adjusted model and using confint(pooled_controls).

18. User prompt (verbatim)

```
Call:
lm(formula = downloads ~ treated + city_size + income_level +
    climate + gym_density, data = ds)

Residuals:
     Min       1Q   Median       3Q      Max
-1.85550 -0.33873  0.02344  0.34143  1.74873

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         7.77258    0.05813 133.716  < 2e-16 ***
treated             1.37177    0.04980  27.544  < 2e-16 ***
city_sizemedium    -0.58929    0.03177 -18.546  < 2e-16 ***
city_sizesmall     -1.00817    0.03557 -28.341  < 2e-16 ***
income_levellow    -0.62283    0.04404 -14.142  < 2e-16 ***
income_levelmiddle -0.28444    0.04079  -6.973 4.49e-12 ***
climatemild        -0.24309    0.03627  -6.701 2.84e-11 ***
climatewarm        -0.11769    0.03282  -3.585 0.000347 ***
gym_densitylow     -0.49417    0.03956 -12.492  < 2e-16 ***
gym_densitymedium  -0.25648    0.03588  -7.147 1.33e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.5093 on 1622 degrees of freedom
Multiple R-squared:  0.7749, Adjusted R-squared:  0.7736
F-statistic: 620.3 on 9 and 1622 DF,  p-value: < 2.2e-16

                       2.5 %      97.5 %
(Intercept)         7.6585646  7.88659039
treated             1.2740899  1.46945937
city_sizemedium    -0.6516125 -0.52696404
city_sizesmall     -1.0779443 -0.93839683
income_levellow    -0.7092180 -0.53644892
income_levelmiddle -0.3644479 -0.20443584
climatemild        -0.3142376 -0.17193623
climatewarm        -0.1820680 -0.05330432
gym_densitylow     -0.5717582 -0.41657954
gym_densitymedium  -0.3268632 -0.18609487
```

Summary of output / discussion: Confirmed Q2 covariate-adjusted estimate 1.37177, CI [1.27409,1.46946], error +0.60177, R-squared 0.7749. The true ATT 0.77 is outside its conventional 95% CI.

19. User prompt (verbatim)

```
The new Pooled ATE is 1.37 with a Confidence Interval of [1.274, 1.469]. The new pooled
ATE is closer to the True ATT (0.6 vs orginal 1.412). The confidence interval also does
not capture True ATT, showing the model is still overestimating treatment effect. Adding
additional covariates helps reduce bias by introducing factors to better define the groups
 of people to analyze behaviors for.
```

Summary of output / discussion: Validated numerical claims; clarified that covariates adjust observable city differences, not individual-person behavioral groups.

---

20. User prompt (verbatim)

```
The new Pooled ATE is 1.37 with a Confidence Interval of [1.274, 1.469]. The new pooled
ATE is closer to the True ATT (0.6 vs orginal 1.412). The confidence interval also does
not capture True ATT.This model is still overestimating the treatment effect. Adding
additional covariates helps reduce bias by introducing charateristics of cities and
factors that could influence app downloads.

does this capture everything stated for q2?
```

Summary of output / discussion: Confirmed Q2 requirements covered; suggested explicitly identifying observable confounding (city size, income, climate, gym density).

---

21. User prompt (verbatim)

```
ds |> filter(downloads %in% c(1, 3))

lm(downloads ~ treated + city_size + income_level +
    climate + gym_density, data = ds) |> summary()


feols(downloads ~ treated + city_size + income_level +
    climate + gym_density, data = ds, cluster = ~store_id)

trying to code wfe1 but getting error
```

Summary of output / discussion: Debugged 1WFE: store_id belongs to webinar grocery example, whereas Billboard uses city; switch to treatedxpost = treated*post, specify fixed effect | city, cluster = ~city. Extraneous filter(downloads %in% c(1,3)) is unnecessary.

---

22. User prompt (verbatim)

```
OLS estimation, Dep. Var.: downloads
Observations: 1,632
Fixed-effects: city: 51
Standard-errors: Clustered (city)
             Estimate Std. Error t value  Pr(>|t|)
treatedxpost 0.941473   0.052819 17.8244 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.397195     Adj. R2: 0.857779
                 Within R2: 0.196138
                 2.5 %   97.5 %
treatedxpost 0.8353818 1.047564

help me understnd how wfe1 works
```

Summary of output / discussion: Explained separate city intercepts, shared treatment slope, control of time-invariant observed/unobserved city attributes. Q3 estimated effect 0.941473, CI [0.835382,1.047564], overestimation ~0.171473.

---

23. User prompt (verbatim)

```
so it's almost like there is a unique model for each unit/city
```

Summary of output / discussion: Affirmed analogy with qualification: one common regression and treatment slope, but unique baseline/intercept for each city.

---

24. User prompt (verbatim)

```
The 1WFE Model allows every city to be evaluated on it's own baseline verses the  shared
baseline pooled regression uses. This means that additional variables become stabilized
within their own cities. Using this model, the ATE is 0.94, which is only 0.17 higher than
 the ATT. Confidence Interval of [0.84, 1.04], which still does not capture True ATT. This
 model is still overestimating treatment effect, but it best performing so far.
```

Summary of output / discussion: Validated overall interpretation; corrected “stabilized” to controls time-invariant city factors, including unobserved attributes; rounded upper CI to 1.05.

---

25. User prompt (verbatim)

```
The 1WFE Model allows every city to be evaluated on it's own baseline verses the  shared
baseline pooled regression uses. This means that additional characteristics are contoled
for each city that remain constant over time Using this model, the ATE is 0.94, which is
only 0.17 higher than the ATT. Confidence Interval of [0.84, 1.05], which still does not
capture True ATT. This model is still overestimating treatment effect, but it best
performing so far.
```

Summary of output / discussion: Confirmed Q3 covers ATE, CI, distinction from pooled/multiple regression, and city-specific time-invariant characteristics. Recommended mentioning unobserved characteristics explicitly.

---

26. User prompt (verbatim)

```
ds <- ds |>
  mutate(treatedxpost = treated * post)

model_2wfe <- feols(
  downloads ~ treatedxpost | city + date,
  data = ds,
  cluster = ~city
)

summary(model_2wfe)
confint(model_2wfe)

OLS estimation, Dep. Var.: downloads

The 2WFE Model allows every city to be evaluated on it's own baseline verses the shared
baseline pooled regression uses. This means that additional characteristics are controlled
 for each city that remain constant over time. This Model evalutes these characteristics
using time as well, to better understand these characteristics. Using this model, the ATE
is 0.82, which is only 0.05 higher than the ATT. Confidence Interval of [0.70, 0.93],
which is the first model to capture ATT

Observations: 1,632
Fixed-effects: city: 51,  date: 32
Standard-errors: Clustered (city)
             Estimate Std. Error t value  Pr(>|t|)
treatedxpost 0.817287   0.056955 14.3496 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.38651     Adj. R2: 0.862632
                Within R2: 0.137865
                 2.5 %    97.5 %
treatedxpost 0.7028887 0.9316855
```

Summary of output / discussion: Confirmed Q4 code with city + date fixed effects and city-clustered SE. Treatment estimate 0.817287, CI [0.702889,0.931686], error +0.047287. Date FE adjust common daily shocks rather than tracking city characteristics over time.

---

27. User prompt (verbatim)

```
The 2WFE Model allows every city to be evaluated on it's own baseline verses the shared
baseline pooled regression uses. This means that additional characteristics are controlled
 for each city that remain constant over time. This model also is fixed on time,
controlling for seasonal changes such as weekends or holidays. Using this model, the ATE
is 0.82, which is only 0.05 higher than the ATT. Confidence Interval of [0.70, 0.93],
which is the first model to capture ATT
```

Summary of output / discussion: Confirmed Q4 interpretation; refined “seasonal changes” to “date-specific influences” and “capture ATT” to “confidence interval includes known ATT”.

---

28. User prompt (verbatim)

```
summarize our models and their outputs
```

Summary of output / discussion: Provided four-model comparison: pooled 2.182 [2.097,2.268]; pooled covariates 1.372 [1.274,1.469]; 1WFE 0.941 [0.835,1.048]; 2WFE 0.817 [0.703,0.932]; true ATT 0.770. 2WFE smallest absolute error ~0.047. Expansion requires positive ROI, not significance alone.

---

29. User prompt (verbatim)

```
Felix, I have completed four regression models to estimate the effect of billboard
advertising on daily app downloads in cities with the Fitlife's billboard treatment.
Geo-experiments are known for having challenges in measuring treatment effect due to
unique characteristics of the day to day life in the world such as weather, median income,
 holidays, ect. Having a baseline understanding of these effects allow us to tailor the
models towards considering these things. To begin, as a baseline, we have the baseline
True ATT of 0.77 additional daily downloads based on historical data. The four models we
tested:

Pooled OLS: Estimated Treatment Effect 2.182, 95% Confidence interval [2.097, 2.268]. This
 model had an overestimation of treatment effect by + 1.412 and the Confidence Interval
did not include ATT.

Pooled OLS + Covarients: Estimated Treatment Effect 1.372, 95% Confidence interval [1.274,
 1.469]. This model had an overestimation of treatment effect by + 0.602 and the
Confidence Interval did not include ATT.

One Way FE: Estimated Treatment Effect 0.941, 95% Confidence interval [0.835, 1.048]. This
 model had an overestimation of treatment effect by + 0.171 and the Confidence Interval
did not include ATT.

Two Way FE: Estimated Treatment Effect 0.817, 95% Confidence interval [0.703, 0.932]. This
 model had an overestimation of treatment effect by + 0.047 and the Confidence Interval
did include ATT.

Based on these four models, the Two Way FE is the best model. In this model,
additional characteristics are controlled for each city that remain constant over time.
This model also is fixed on time, controlling for seasonal changes such as weekends or
holidays. Because of this feature, it is able to provide us the best true estimated
average treatment effect. With this, there is supporting evidence that there are
additional downloads with the billboard marketing efforts and this model should be
extended to the control cities if there is a positive ROI with the number of true
downloads compared against the marketing expense.
```

Summary of output / discussion: Q5 feedback: add experiment design details, explicitly discuss pretreatment treated-control gaps, clarify tau yields known true ATT (not historical data), retain model comparison and conditional ROI expansion recommendation.

---

30. User prompt (verbatim)

```
Felix, I have completed four regression models to estimate the effect of billboard
```

```
advertising on daily app downloads in cities with the Fitlife's billboard treatment.
Geo-experiments are known for having challenges in measuring treatment effect due to
unique observed characteristics of the day to day life in the world such as weather, gymn
density, population median income, holidays, ect. There are also unobersved differences
between cities that could influence downloades. Some models are not able to distinguish
the differences that were pre-exisiting and those caused by the treatment.

In this experiment there were 51 cities, 9 treated, 42 control. May 1-14 is the
Pre-Treatment period, May 15-June 1 is the Treatment period. Daily app downloads is
measured for both treatment and control groups. Cities are selected based on
characteristics rather than explicitly randomized. As a baseline, we have the baseline
True ATT of 0.77 additional daily downloads.

The four models we tested:

Pooled OLS: Estimated Treatment Effect 2.182, 95% Confidence interval [2.097, 2.268]. This
 model had an overestimation of treatment effect by + 1.412 and the Confidence Interval
did not include ATT.

Pooled OLS + Covarients: Estimated Treatment Effect 1.372, 95% Confidence interval [1.274,
 1.469]. This model had an overestimation of treatment effect by + 0.602 and the
Confidence Interval did not include ATT.

One Way FE: Estimated Treatment Effect 0.941, 95% Confidence interval [0.835, 1.048]. This
 model had an overestimation of treatment effect by + 0.171 and the Confidence Interval
did not include ATT.

Two Way FE: Estimated Treatment Effect 0.817, 95% Confidence interval [0.703, 0.932]. This
 model had an overestimation of treatment effect by + 0.047 and the Confidence Interval
did include ATT.

Based on these four models, the Two Way FE is the best model. In this model,
additional characteristics are controlled for each city that remain constant over time.
This model also is fixed on time, controlling for seasonal changes such as weekends or
holidays. Because of this feature, it is able to provide us the best true estimated
average treatment effect. With this, there is supporting evidence that there are
additional downloads with the billboard marketing efforts and this model should be
extended to the control cities if there is a positive ROI with the number of true
downloads compared against the marketing expense.
```

Summary of output / discussion: Final Q5 review: all four requirements addressed. Suggested direct reference to pre-campaign download differences, clarifying that ATT was computed from tau and not observed historical ground truth, date fixed effects capture common daily influences, and ROI requires campaign cost and value/conversion information.

---

31. User prompt (verbatim)

```
please render a PDF of our session
```

Summary of output / discussion: An initial 3-page condensed PDF summary was created, but the user requested a more complete verbatim prompt log instead.

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32. User prompt (verbatim)

```
no, this needs word for word prompts and summary of the outputs.
```

Summary of output / discussion: This revised PDF reproduces the user text prompts in this session and follows each with a concise summary of the assistant response or model output. Screenshot-only content is described, because it is not a typed text prompt.
